



**Northern Beaches Secondary College
Manly Selective Campus**

**2010
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Board-approved calculators and templates may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1- 7
- All questions are of equal value

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Marks

Question 1 (Answer in a separate booklet)

(12)

a) Solve for x in the following. $\frac{2x}{x-2} \leq 5$ (2)

b) Determine the size of the acute angle formed by the intersection of the lines
 $2x + y = 3$ and $3x - 5y = 10$. (2)

c) How many ways could the letters in the word LEVEL be arranged if the
two letter E's are together? (2)

d) Show that $(2x + 1)$ is a factor of $P(x) = 2x^3 - x^2 - 13x - 6$. (1)

Hence find a second expression which is a factor of $P(x)$ (1)

e) Find $\int \sin^2 3x \cdot dx$ (2)

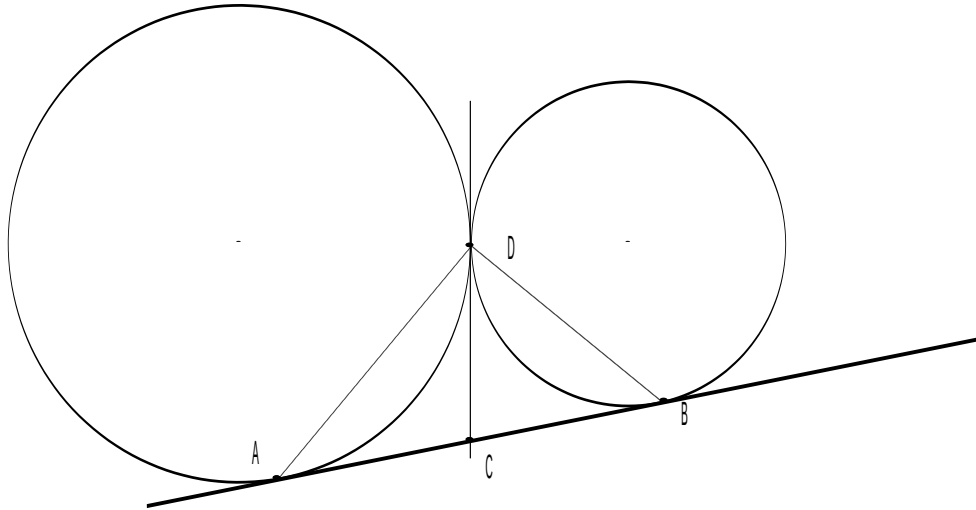
f) Determine the exact value of $\sin 165^\circ$. (2)

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Marks
(12)

Question 2 (Answer in a separate booklet)

a)



Two circles have a common tangent CD and meet externally at D.
AB is another common tangent with A and B being points of tangency.
AB and CD intersect at C.

- (i) Give reasons to explain why $AC = BC$ (1)
- (ii) Prove that $\angle ADB = 90^\circ$ (3)
- b) One quarter of all jellybeans in a mixture of jellybeans are black. A random sample of ten jellybeans is taken. Find the probability that the sample contains
- (i) no black jellybeans (1)
- (ii) two black jellybeans (2)
- c) Prove by mathematical induction for all positive integers n that $13 \times 6^n - 3$ is divisible by 5 (3)
- d) Find the constant term in the expansion of
- $$\left(x - \frac{1}{2x^2}\right)^{15}$$
- (2)

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Question 3 (Answer in a separate booklet)

(12)

- a) From the top of a lighthouse 55 metres high, a boat is observed due south at an angle of depression of $18^{\circ}50'$. The boat is sailing due east and after 5 minutes, the angle of depression is found to be $12^{\circ}12'$.

Calculate the speed of the boat in km/hr to the nearest km.

(4)

- b) The equation $x^3 - 3x^2 + 9 = 0$ has a single root. In an effort to determine its value, Dani takes $x = 3$ as her first approximation for that root and uses Newton's method twice.

- (i) Interpret the value you have calculated based on Dani's choice for the value of the root.

(2)

- (ii) What would be a better starting approximation?
Give reasons for your answer.

(2)

- c) (i) Show that $\frac{u}{u+1} = 1 - \frac{1}{u+1}$

(1)

- (ii) Hence find $\int \frac{1}{1 + \sqrt{x}} dx$ using the substitution $x = u^2$ ($u \geq 0$).

(3)

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(12)

Question 4 (Answer in a separate booklet)

(a) Four girls and two boys are to be seated **around** a table.

(i) How many seating arrangements are possible without restriction? (1)

(ii) Determine the probability that if the seating is arranged randomly that the two boys do **NOT** sit together. (3)

(b) A monic polynomial, when divided by $x^2 - 4$ leaves a remainder of $7x - 4$ and, when divided by x , there is a remainder of -8 .

Write out the polynomial in the form $P(x) = ax^3 + bx^2 + cx + d$ (4)

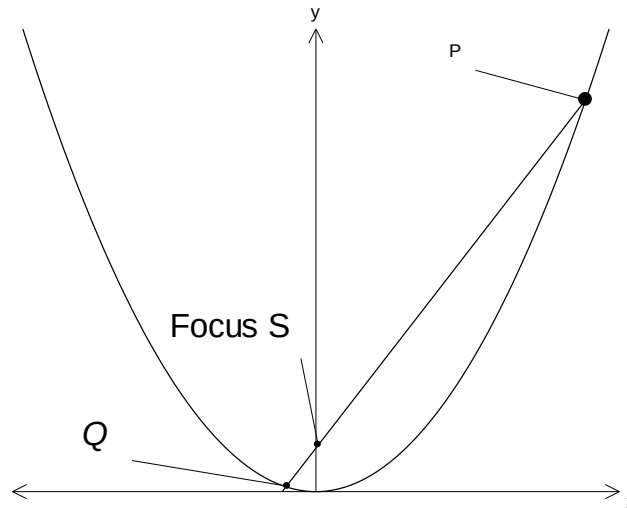
(c) (i) Show that $\cos\left(\frac{\pi}{2} + A\right) = -\sin A$ for all A (1)

(ii) Taking $A = 5\theta$, write down the value of B such that $-\sin 5\theta = \cos B$.
Hence find the least value of θ , where $0 \leq \theta \leq 2\pi$,
such that $\sin 5\theta + \cos 8\theta = 0$ (3)

Question 5 (Answer in a separate booklet)

(12)

(a)



PQ is a chord on the parabola $x^2 = 4ay$ drawn above. If the parametric coordinates of P and Q are respectively $(2ap, ap^2)$ and $(2aq, aq^2)$ then show that:

(i) the equation of the chord PQ is $y = \frac{p+q}{2}x - apq$ (3)

(ii) if PQ is a focal chord then $pq = -1$ (1)

(iii) the length of the focal chord PQ is $a\left(p + \frac{1}{p}\right)^2$ (2)

(b) (i) Simplify $\cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right)\right)$ (1)

(ii) Find $\frac{d}{dx}\left(\tan^{-1}\left(\frac{x}{3}\right)\right)$ in simplest form (2)

(iii) Evaluate $\int_{0.25}^{0.5} \frac{dx}{\sqrt{1-4x^2}}$ (3)

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Question 6 (Answer in a separate booklet)

(12)

(a) Assume that the rate at which a body warms in air is proportional to the difference between its temperature T and the constant room temperature A of the surrounding air. The rate can be expressed by the differential equation $\frac{dT}{dt} = k(T - A)$ where t is the time in minutes and k is a constant.

(i) Show that $T = A + C.e^{kt}$, where C is a constant, is a solution of the differential equation. (1)

(ii) A cooled body warms from 5°C to 10°C in 20 minutes. The air temperature around the body is 25°C .

Using this information, show that the value for $k = 0.05 \times \ln 0.75$ (2)

(iii) Determine the temperature of object of the body after a further 40 minutes have elapsed. Give your answer to the nearest degree. (2)

(iv) By referring to the equation for T , explain the behaviour of T as t becomes larger. (1)

(b) Show that $\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$ (1)

(c) The acceleration of a particle moving in a straight line is given by $\frac{d^2x}{dt^2} = 2x - 3$

where x is the displacement, in metres, from the origin O and t is the time in seconds. Initially the particle is at rest $x = 4$.

(i) If the velocity of the particle is $v \text{ ms}^{-1}$, determine an expression for v^2 as a function of x . (2)

(ii) Show that the particle does not pass through the origin. (1)

(iii) Determine the position of the particle when $v = 10$. Justify your answer. (2)

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Question 7 (Answer in a separate booklet)

(12)

a) If $(1 + x)^7 = 1 + a_1x + a_2x^2 + a_3x^3 + \dots + a_7x^7$ and

$$(1 - x)^{10} = 1 + b_1x + b_2x^2 + b_3x^3 + b_4x^4 + \dots + b_{10}x^{10}$$

Find:

(i) the value of $a_4b_7 - b_4a_7$ (2)

(ii) the greatest value of a_nb_n where $1 \leq n \leq 7$. (2)

b) A particle moving in simple harmonic motion has a period of $\frac{\pi}{3}$ sec and a maximum velocity of 2 metres/sec. Find the amplitude of the motion and its displacement after $\frac{\pi}{6}$ seconds. (2)

c) The displacement x metres of a particle moving in simple harmonic motion is given by $x = 4 \sin \pi t$ where t is the time expressed in seconds.

(i) What is the period of oscillation? (1)

(ii) What is the speed of the particle as it moves through the centre of motion? (2)

(iii) Show that the acceleration of the particle is proportional to the displacement from the equilibrium position (centre of motion). (1)

(iv) What is the velocity of the particle at the position $x = 3$? (2)

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

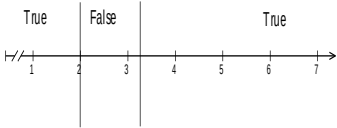
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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Question 1

(a)	$\frac{2x}{x-2} \leq 5 \quad \therefore x \neq 2$ $\frac{2x}{x-2} \leq 5$ $2x \leq 5x - 10$ $10 \leq 3x$ $\frac{10}{3} \leq x$  $\therefore x < 2 \text{ or } \frac{10}{3} \leq x$	2 marks – correct answer 1 mark – incorrect inequality for 2
(b)	$2x + y = 3 \quad m = -2$ $3x - 5y = 10 \quad m = \frac{3}{5}$ $\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 \times m_2} \right $ $= \left \frac{-2 - \frac{3}{5}}{1 + \frac{3}{5} \times -2} \right $ $= \left \frac{-\frac{13}{5} \times \frac{-5}{1}}{1} \right $ $\theta = \tan^{-1} 13$ $= 85^\circ 36'$	2 marks – correct answer 1 mark – correct substitution into correct formula
(c)	LEVEL $\text{No. of arrangements} = \frac{4!}{2!} = 12$	2 marks – correct answer 1 mark – incorrect denominator
(d) - i	$P(x) = 2x^3 - x^2 - 13x - 6 \quad x = -\frac{1}{2}$ $P\left(-\frac{1}{2}\right) = 2\left(-\frac{1}{2}\right)^3 - \left(-\frac{1}{2}\right)^2 - 13\left(-\frac{1}{2}\right) - 6$ $= 0$	1 marks - Correct demonstration.

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Question 1 (continued)

(d)ii	Factors of 6 = $\pm(1,2,3,6)$ $P(1) = -18$ $P(2) = -20$ $P(3) = 0$ $P(-2) = 0$ Therefore $(x-3)$ and $(x+2)$ are both factors	1 mark – correct answer
(e)	$\int \sin^2 3x \cdot dx$ $= \frac{1}{2} \int 1 - \cos 6x \cdot dx$ $= \frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + C$	2 marks – correct answer 1 mark – correct substitution.
(f)	$\sin 165^\circ = \sin(120 + 45)$ $= \sin 120 \cdot \cos 45 + \sin 45 \cdot \cos 120$ $= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \times -\frac{1}{2}$ $= \frac{\sqrt{3} - 1}{2\sqrt{2}}$ $= \frac{\sqrt{6} - \sqrt{2}}{4}$	2 marks – correct answer 1 mark – correct expansion for compound angle.

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Question 2

Q2(a)	AC=DC= BC (tangents drawn from an external point are equal)	1- Correct reason
(i)	As CA=CB=CD then C is the centre of a circle with A,B and D on its circumference. A,C and B are collinear thus $\angle ACB = 180^\circ$ AB is a diameter $\angle ADB = \frac{1}{2} \times \angle ACB = 90^\circ$ (Angle in a semi-circle is a right angle)	3- Correct reasons given for all steps. 2- All but one reason given 1- Determining that C is centre of circle
(b) (i)	$\left(\frac{1}{4} + \frac{3}{4}\right)^{10}$ $P(\text{no black}) = \left(\frac{3}{4}\right)^{10} \cong 0.0563$	1- correct answer
(b)(ii)	$\left[\begin{matrix} 10 \\ 2 \end{matrix} \right] \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^8 \cong 0.2816$	2-Correct term of binomial and answer 1-Correct term
(c)	* Test true for $n = 1$ $LHS = 13 \times 6 - 3 = 75$ $RHS = 5 \times 15$ Thus true for $n=1$ * Assume true for $n = k$ Thus $13 \times 6^k - 3 = 5p$ (p is a positive integer) * Prove true for $n=k+1$ That is $LHS = 13 \times 6^{k+1} - 3 = 5q = RHS$ Proof: $LHS =$ $6 \times 13 \times 6^k - 3 = 6(5p + 3) - 3 = 30p + 15$ $= 5(6p + 3) = 5q$ * Tested true for $n=1$, proved true for $n=k+1$, thus true for all n	3- Complete proof with , testing, assumption, proof and conclusion 2- setting out correct with assumption used at proof stage but incomplete thereafter 1- Test and assumption correct
(d)	General term is: $\left[\begin{matrix} 15 \\ k \end{matrix} \right] x^{15-k} \left(\frac{-1}{2x^2}\right)^k = \left[\begin{matrix} 15 \\ k \end{matrix} \right] \left(-\frac{1}{2}\right)^k x^{15-3k}$ $\therefore 15-3k = 0 \therefore k = 5$ Term is: $\left[\begin{matrix} 15 \\ 5 \end{matrix} \right] \left(-\frac{1}{2}\right)^5 = -93\frac{27}{32}$	2- Correct term identified and value found 1- strong evidence of correct term being identified

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Question 3

(a)	$BS = \frac{55}{\tan 18^{\circ}50'}$ and $BE = \frac{55}{\tan 12^{\circ}12'}$ $\therefore SE = \sqrt{\frac{55^2}{\tan^2 12^{\circ}12'} - \frac{55^2}{\tan^2 18^{\circ}50'}}$ = 196.75 m $\therefore$ in one hour, distance = 2361 m $\therefore$ speed = 2 km/hr <i>Note: Confusion over where an angle of depression is located. Clear need to draw a 3D diagram properly so you know where the right angle is and you don't think 2D (and hence just subtract).</i>	4 marks – correct speed 3 marks – correct distance in 1 hour 2 marks – correct distance between positions 1 mark – correct distances substituted into Pythagoras
(b) (i)	$f(x) = x^3 - 3x^2 + 9$ $f'(x) = 3x^2 - 6x$ $f(3) = 9 \text{ and } f'(3) = 9$ $\therefore x_1 = 3 - \frac{9}{9} = 2$ Dani used the method twice ... so $f(2) = 5 \text{ and } f'(2) = 0$ $\therefore x_2 = 2 - \frac{5}{0} \text{ which is indeterminate}$ so $x = 2$ must be a stationary point so Dani cannot obtain a better approximation with this starting point Indeed the derivative shows SP at $x = 0$ and $x = 2$.	2 marks – can't proceed because gradient = 0 and so tangent being horizontal cannot cut through the x-axis. 1 mark – at $x = 2$ there is a stationary point. NOTE: Surely by now the number $\frac{5}{0}$ looks different and is $\neq 0!!$ It is easy to identify stationary points from the derivative and hence sketch a rough curve to assist this part and the next.
(ii)	A better starting point tries to avoid the SP. If Dani chose $x = 4$, $f(4) = 25$ which is a big value and so a long way from the root. Go the other way. Clearly $f(0) = 9$. So try $x = -1$. $f(-1) = 5$. That is probably OK. It is a reasonably small value for the ordinate. NOTE: Be careful of making statements in a rush and not writing down what you really mean. One statement, repeated several times in different forms, was “the first estimate should be a negative number greater than 3”.	2 marks – estimate of new root with appropriate reason. 1 mark – a better estimate cited.
(c) (i)	$\text{RHS} = 1 - \frac{1}{u+1}$ $= \frac{u+1-1}{u+1}$ $= \frac{u}{u+1} = \text{LHS}$	1 mark – correct proof with correct setting out NOTE: Proof or “Show” answer should start with LHS or RHS

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(ii)	$x = u^2$ $dx = 2u \, du$ $\therefore \int \frac{2u \, du}{1+u} = 2 \int \left[1 - \frac{1}{u+1} \right] du$ $= 2[u - \ln(u+1)] + c$ $= 2[\sqrt{x} - \ln(\sqrt{x} + 1)] + c$	<i>3 marks – correct integral in terms of x.</i> <i>2 marks – correct integral in terms of u</i> <i>1 mark – correct substitution</i>
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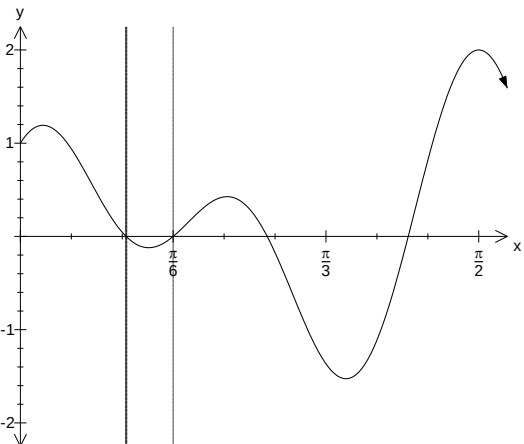
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Question 4

(a) (i)	$5! = 120$	<i>1 mark – correct answer</i>
(ii)	Number = Total – No. (Boys sit together) $n = 5! - 4! \times 2!$ $= 72$ $\text{Prob} = \frac{72}{120} = \frac{3}{5}$	<i>3 marks - Boys seated together calculated, then boys NOT together calculated then probability found.</i> <i>2 marks - Boys seated together calculated, then boys NOT together calculated</i> <i>1 marks - An reasonable attempt at the calculation</i>
(b)	$P(x) = ax^3 + bx^2 + cx + d$ Polynomial is monic so $a = 1$ If divided by x $P(0) = d = -8$ $P(x) = x^3 + bx^2 + cx - 8$ $P(2) = 8 + 4b + 2c - 8 = 14 - 4$ $\therefore 4b + 2c = 10$ $P(-2) = -8 + 4b - 2c - 8 = -14 - 4$ $4b - 2c = -2$ $8b = 8 \therefore b = 1$ $c = 3$ $\therefore x^3 + x^2 + 3x - 8$	<i>4 marks → 1 mark for each coefficient</i>

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Question 4 (c) (continued)

$LHS = \cos\left(\frac{\pi}{2} + A\right)$ $= \cos \frac{\pi}{2} \cdot \cos A - \sin \frac{\pi}{2} \sin A$ $= 0 - 1 \times \sin A$ $= -\sin A$ $= RHS$	1 mark – correct solution fully demonstrated.
$B = \frac{\pi}{2} + 5\theta$ $-\sin 5\theta = \cos\left(\frac{\pi}{2} + 5\theta\right) = \cos 8\theta$ In 1st and 4th quad: $\frac{\pi}{2} + 5\theta = 8\theta$ $\theta = \frac{\pi}{6}$ In 2nd and 3rd quads, $\cos(-\theta) = \cos \theta$ so reverse one side of the equation $\frac{\pi}{2} + 5\theta = -8\theta$ $13\theta = -\frac{\pi}{2} = \frac{3\pi}{2} \text{ (as no soln in the given range for -)}$ $13\theta = \frac{3\pi}{2}$ $\theta = \frac{3\pi}{26}$ Of these 2 solns, $\frac{3\pi}{26} < \frac{\pi}{6}$	3 marks - correct answer identified 2 marks - correct answer for $\frac{\pi}{6}$ 1 mark – correct expression for B A graph helps to see the context of this question (not expected to be drawn – just for background): 

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Question 5

5(a) (i)	$m(\text{gradient of chord}) = \frac{ap^2 - aq^2}{2aq - 2ap} = \frac{a(p-q)(p+q)}{2a(p-q)}$ $= \frac{p+q}{2}$ equation of chord: $y - ap^2 = \left(\frac{p+q}{2}\right)(x - 2ap)$ $y = \left(\frac{p+q}{2}\right)x - \left(\frac{p+q}{2}\right)2ap^2 + ap^2$ $y = \left(\frac{p+q}{2}\right)x - ap^2 - apq + ap^2$ $y = \left(\frac{p+q}{2}\right)x - apq$	3- Correct equation with correct gradient found 2- Correct gradient from correct factoring found substituted into equation format 1- Correct gradient from correct factoring found
(ii)	If focal chord: $a = -apq$ $-l = pq$	1- Correctly demonstrated
(iii)	Use lengths to directrix: $SP = PD_1 = ap^2 + a = a(p^2 + 1)$ $SQ = QD_2 = aq^2 + a = a(q^2 + 1)$ $\therefore \overrightarrow{PQ} = a(p^2 + 1 + q^2 + 1)$ but $q = -\frac{1}{p}$ $= a\left(p^2 + 2 + \frac{1}{p^2}\right) = a\left(p + \frac{1}{p}\right)^2$	2- correct length found from appropriate calculation 1- Correct initial statement of length with $q = -\frac{1}{p}$ used at some stage
(b) (i)	$\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$	1- Correct value
(ii)	$\frac{1}{1 + \left(\frac{x}{3}\right)^2} \times \frac{1}{3} = \frac{1}{\frac{9+x^2}{9}} \times \frac{1}{3} = \frac{3}{9+x^2}$	2- Correct derivative process then simplified 1- Correct derivative but not fully simplified
(iii)	$\frac{1}{2} \int_{0.25}^{0.5} \frac{2x dx}{\sqrt{1 - (2x)^2}} = \left[\frac{1}{2} \sin^{-1}(2x) \right]_{0.25}^{0.5}$ $= \left[\frac{1}{2} \times \frac{\pi}{2} - \frac{1}{2} \times \frac{\pi}{6} \right]$ $= \frac{\pi}{6}$	3- Correct integral found then correct exact value 2- Correct integral found 2 0.5236 2- $\sin^{-1}(2x)$ then $\frac{\pi}{3}$ 1- $\sin^{-1}(2x)$ sighted
Comments: (a)(i)(ii) done well by most students (iii) Students attempted to use distance formula which proved most difficult. (b) (i) Too many said the easy wrong answer $\frac{4\pi}{3}$ (ii) (iii) generally answered well		

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Question 6

6(a)(i)	$T = A + Ce^{kt}$ $\frac{dT}{dt} = k \cdot Ce^{kt}$ $= kA + k \cdot Ce^{kt} - kA$ $= k(A + Ce^{kt}) - kA$ $= kT - kA$ $= k(T - A)$ $= RHS$	1 mark – correctly demonstrated.
(ii)	$T = 5 \quad t = 0 \quad A = 25$ $T = A + Ce^{kt}$ $5 = 25 + C \cdot e^0$ $-20 = C$ $T = 10 \quad t = 20 \quad A = 25$ $10 = 25 - 20 \cdot e^{k \times 20}$ $\frac{15}{20} = e^{20k}$ $\frac{1}{20} \ln\left(\frac{3}{4}\right) = k$	2 marks – correct explanation 1 mark – correct value for C
(iii)	$T = x \quad t = 60 \quad A = 25$ $T = A + Ce^{kt}$ $x = 25 - 20 \cdot e^{60 \times \frac{1}{20} \ln \frac{3}{4}}$ $x = 16.56$ Therefore temperature of particle is 17^0 – nearest degree.	2 marks – correct answer 1 mark – a correct answer using incorrect time.

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iv	$\text{As } t \rightarrow \infty \quad \text{then } e^{kt} \rightarrow e^{-\infty} \rightarrow 0$ $\therefore \lim_{t \rightarrow \infty} T$ $= \lim_{t \rightarrow \infty} 25 - e^{kt}$ $= 25 - 0$ $= 25$ Therefore T approaches 25°C	1 mark – fully and correctly demonstrated with indication that $e^{kt} \rightarrow e^{-\infty} \rightarrow 0$
6b-(i)	$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ $RHS = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ $= \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \times \frac{dv}{dx}$ $= v \times \frac{dv}{dx}$ $= \frac{dx}{dt} \times \frac{dv}{dx}$ $= \frac{dv}{dt}$ $= \ddot{x}$ $= RHS$	1 mark – correct explanation with all steps fully demonstrated.
6c (i)	$\frac{d^2x}{dt^2} = 2x - 3$ $\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2x - 3$ $\frac{1}{2} v^2 = x^2 - 3x + \frac{C}{2}$ $v^2 = 2x^2 - 6x + C$ When $v=0$, $x=4$ $0 = 2 \times (4)^2 - 6 \times 4 + C$ $C = -8$ $v^2 = 2x^2 - 6x - 8$	2 marks – correct answer fully explained. 1 mark - incorrect value for C.
c (ii)	Origin $x = 0 \quad v^2 = -8$ As this is not possible therefore particle does not pass through the origin.	1 mark – correct explanation.

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c(iii)	when $v = 10$ $100 = 2x^2 - 6x - 8$ $0 = x^2 - 3x - 54$ $0 = (x - 9)(x + 6)$ $x = 9 \text{ or } -6$ $x = 9$ as particle starts at 4 and does not pass through the origin.	2 marks – correct answer with correct explanation. 1 mark – failure to explain correctly the exclusion of $x = -6$.
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Comments:

6a-iv – Brevity is not an asset in Proof questions – too many students failed to adequately show $e^{kt} \rightarrow e^{-\infty} \rightarrow 0$ as part of the required answer.

6b - A number of students introducing steps without previous justification and/or introducing assumptions/statements in incorrect manner.

6 c-i A number of students left out $\frac{1}{2}$ in $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \ddot{x}$

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Question 7

(a) (i)	$a_4 b_7 - b_4 a_7 = {}^7C_4({}^{10}C_7) - {}^{10}C_4 {}^7C_7$ $= -4410$	2 marks – correct value determined 1 mark – correct identification of coefficients
(ii)	The n subscripts are the same in both terms. $\frac{{}^7C_n {}^{10}C_n}{{}^7C_{n-1} {}^{10}C_{n-1}} > 1$ After expanding and cancelling factorials we get $\frac{8-k}{k} \times \frac{11-k}{k} > 1$ $88 - 19k + k^2 > k^2$ $\frac{88}{19} > k$ $\therefore k = 4$ $\therefore {}^7C_4 {}^{10}C_4 = 7350$	2 marks – correct greatest value 1 mark – correct approach to determining k
(b)	Period = $\frac{\pi}{3}$ $\therefore \frac{2\pi}{n} = \frac{\pi}{3} \therefore n = 6$ $v^2 = n^2(a^2 - x^2)$ $4 = 36(a^2 - 0)$ Amplitude = $a = \frac{1}{3}$ $\therefore x = \frac{1}{3} \cos(6t)$ at $t = \frac{\pi}{6}$ $x = -\frac{1}{3}$ which is $\frac{1}{2}$ a wavelength away from the starting point If using a sin curve the particle is back at the origin so again half a wavelength different	2 marks – correct amplitude and displacement 1 mark – correct amplitude

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Question 7 (continued)

(c) (i)	$n = \pi$ $\text{period} = \frac{2\pi}{n} = \frac{2\pi}{\pi} = 2 \text{ seconds}$	1 mark – correct period
(ii)	The particle starts at the origin at $t = 0$ so it returns after half the period – i.e. at $t = 1$. $x = 4 \sin \pi t$ $\dot{x} = 4\pi \cos \pi t$ at $t = 1$ $\dot{x} = 4\pi \cos \pi = -4\pi$	2 marks – correct velocity determined 1 mark – correct approach but minor arithmetic error
(iii)	$\dot{x} = 4\pi \cos \pi t$ (from (ii)) $\ddot{x} = -4\pi^2 \sin \pi t$ $= -\pi^2 x$ $\therefore$ acceleration is negatively proportional to the displacement from origin <i>NOTE: You must set out questions of SHOW, PROVE etc fully (as noted previously). If you do not, you will be heavily penalized. So do not short cut!! Ever!!</i>	1 mark – correct demonstration
(iv)	$v^2 = n^2 (a^2 - x^2)$ $= \pi^2 (4^2 - 3^2)$ $v = \sqrt{7} \pi$	2 marks – correct answer 1 mark – correct approach using v^2